ch8 Work Energy and Power

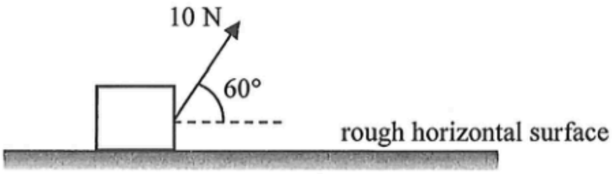
2012 Q9

9. An object of mass 0.5 kg is raised vertically from the ground by a motor. The object is raised 2.5 m in 1.5 s with uniform speed. Estimate the output power of the motor. Neglect air resistance. ($g = 9.81\ \mathrm{m\ s^{-2}}$)

A. 5.5 W
B. 8.2 W
C. 11.0 W
D. 16.4 W

2025 Q5

5. A block, initially at rest, is pulled along a horizontal surface by a force of 10 N making an angle of 60° with the horizontal. There is a constant friction of 2 N between the block and the horizontal surface. What is the gain in kinetic energy of the block after moving 5 m ?



A. 33 J
B. 25 J
C. 15 J
D. 10 J

2024 Q7

7. In an experiment, a horizontal force F acts on a block placed on a rough horizontal surface as shown in Figure (a). Figure (b) shows the variation of the block's kinetic energy E_k with the distance s it travelled. Neglect air resistance and assume frictional force unchanged.

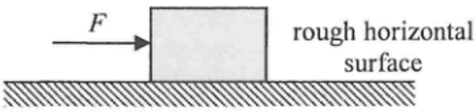


Figure (a)

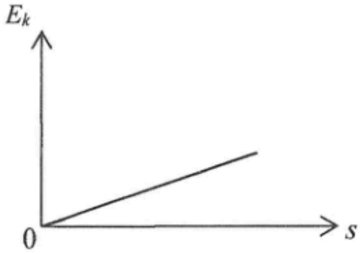
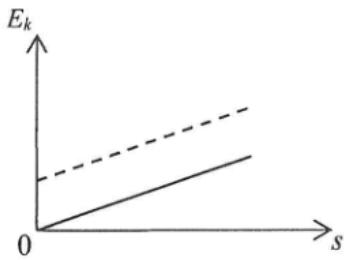


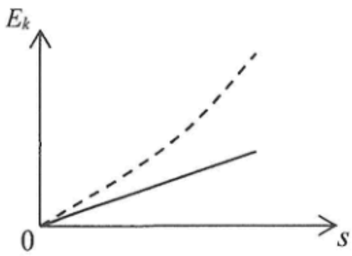
Figure (b)

If the experiment is repeated with force F doubled, which of the following graphs represents the expected result (in dotted line) ?

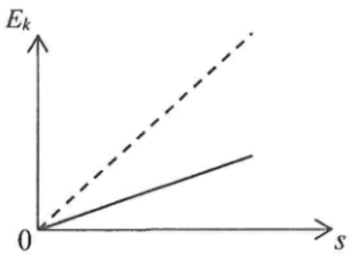
A.



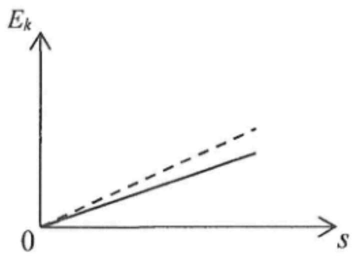
B.



C.

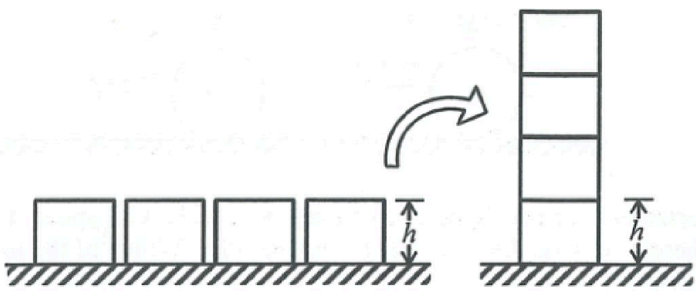


D.



2019 Q9

9.

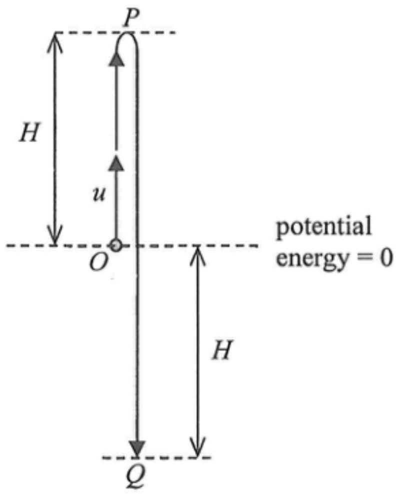


Four identical uniform blocks, each of mass m and height h, are first placed on a horizontal table. If the blocks are stacked on top of one another as shown, what is the minimum work done ?

A. $8\,mgh$
B. $6\,mgh$
C. $4\,mgh$
D. $3\,mgh$

2025 Q8

8. A small ball of mass m is thrown vertically upward from point O with an initial speed u. It reaches the highest point P and then falls to point Q such that $OP = OQ = H$. The ball's potential energy at point O is taken to be zero.

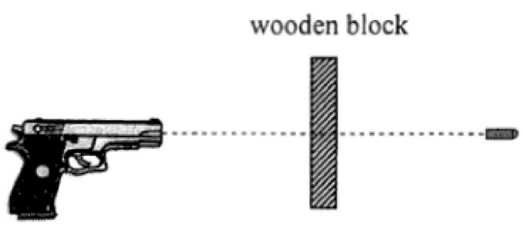


What are the kinetic energy and potential energy of the ball at point Q ? Neglect air resistance.

	kinetic energy	**potential energy**
A.	mu^2	$-mgH$
B.	mu^2	mgH
C.	$\frac{1}{2}mu^2$	$-mgH$
D.	$\frac{1}{2}mu^2$	mgH

2013 Q12

12.

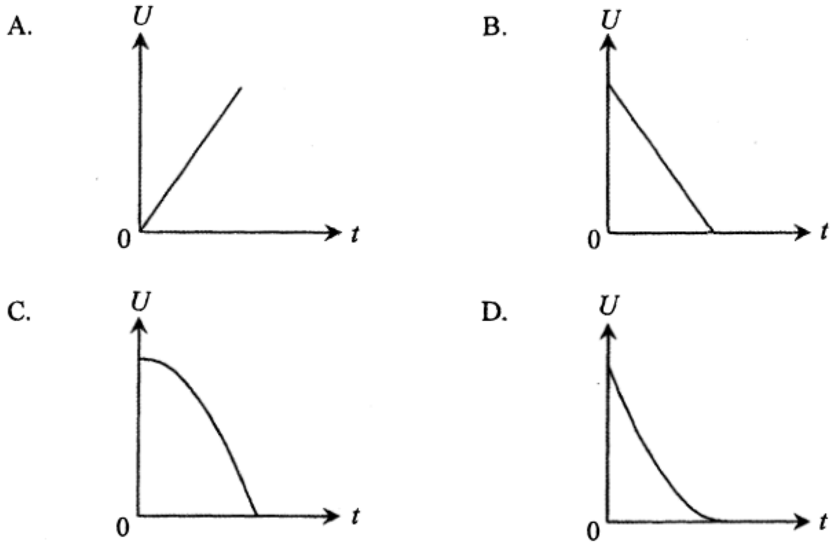


A bullet of mass 50 g is fired from a gun with a speed of 400 m s^{-1} and passes right through a fixed wooden block of 6 cm thickness as shown. Find the average resistive force acting on the bullet due to the block if it emerges with a speed of 250 m s^{-1}. Neglect air resistance and the effects of gravity.

A. 4.06×10^4 N
B. 1.02×10^4 N
C. 125 N
D. Answer cannot be found as the time of travel of the bullet within the block is not known.

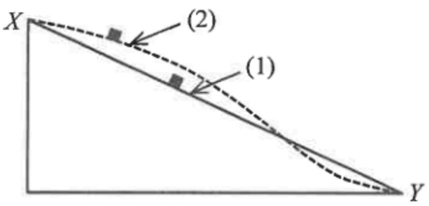
2016 Q8

8. An object at a certain height falls freely from rest under gravity. Which graph correctly shows the variation of its gravitational potential energy U with time t ? Neglect air resistance and take $U = 0$ at the ground.



2022 Q11

11. In the figure below, the straight path (1) and the curved path (2) in a vertical plane are smooth. A small block slides from rest at X along the respective paths to Y.



Which of the following about the speed of the block at Y and the time taken to reach Y is correct ? Neglect air resistance.

	speed at Y	**time taken to reach Y**
A.	the same	different
B.	the same	the same
C.	different	the same
D.	different	different

2014 Q6

6. Two small identical blocks slide down from rest on smooth incline planes from the same height H as shown in Figure (1) and Figure (2) below. Their respective speeds at the bottom of the incline planes are denoted by v_1 and v_2 and the respective times taken to reach the bottom are t_1 and t_2. Which of the following is correct ? Neglect air resistance.

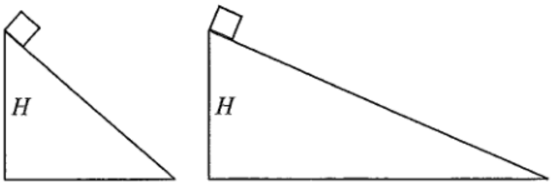


Figure (1) Figure (2)

A. $v_1 > v_2$ and $t_1 = t_2$

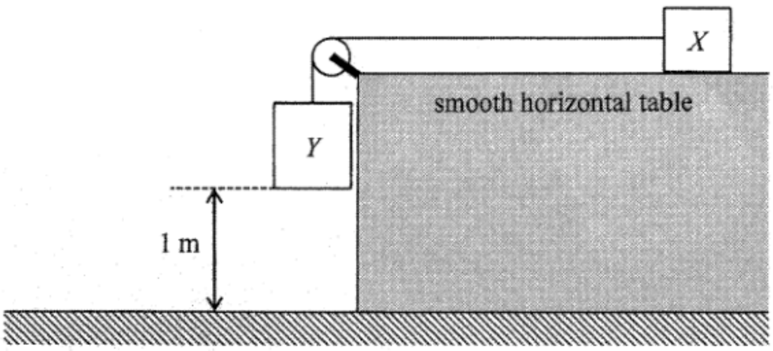
B. $v_1 > v_2$ and $t_1 < t_2$

C. $v_1 = v_2$ and $t_1 = t_2$

D. $v_1 = v_2$ and $t_1 < t_2$

2017 Q10

10. Blocks X and Y are connected by a light inextensible string passing over a fixed frictionless light pulley as shown. The mass of X and Y are 0.5 kg and 1 kg respectively. Initially, Y is 1 m above the ground and the string is taut. The system is then released from rest.

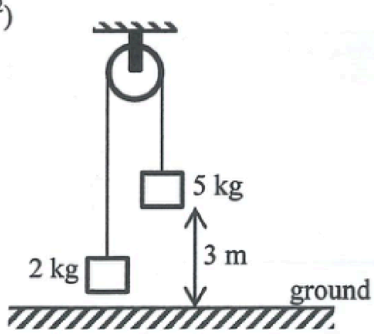


What is the speed of Y just before it reaches the ground ? (Take $g = 9.81 \text{ m s}^{-2}$)

A. 3.62 m s^{-1}
B. 4.43 m s^{-1}
C. 6.26 m s^{-1}
D. 9.81 m s^{-1}

2019 Q12

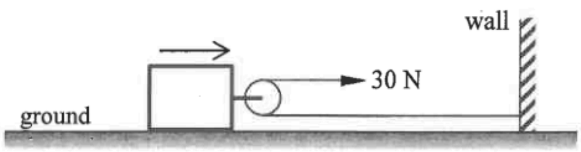
12. Two blocks of respective masses 2 kg and 5 kg are connected by a light inextensible string which passes over a smooth fixed light pulley as shown. The system is released from rest when the 5-kg block is 3 m above the ground. What is the speed of the 5-kg block just when reaching the ground ? Neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)



A. 5.0 m s^{-1}
B. 6.0 m s^{-1}
C. 6.5 m s^{-1}
D. 7.7 m s^{-1}

2022 Q10

10.

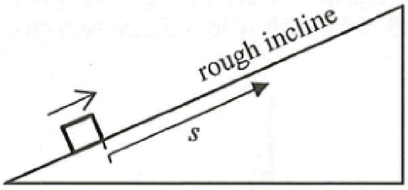


On a horizontal ground there is a block attached with a smooth light pulley as shown. A light and inextensible horizontal string, with one end fixed to a wall, passes the pulley. A man exerts a horizontal force of 30 N at the other end of the string and pulls a distance of 4 m. Find the work done by the man if the ground exerts a frictional force of 10 N on the block.

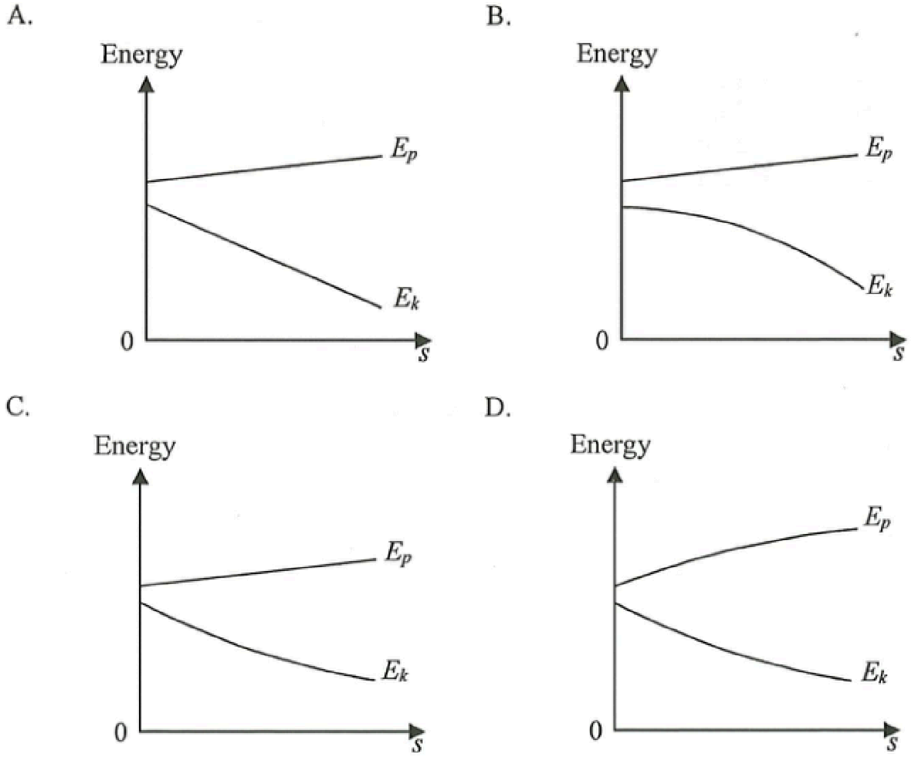
A. 20 J
B. 80 J
C. 100 J
D. 120 J

2023 Q10

10. A block is projected upwards along a **rough** incline which has a constant frictional force acting on the block.

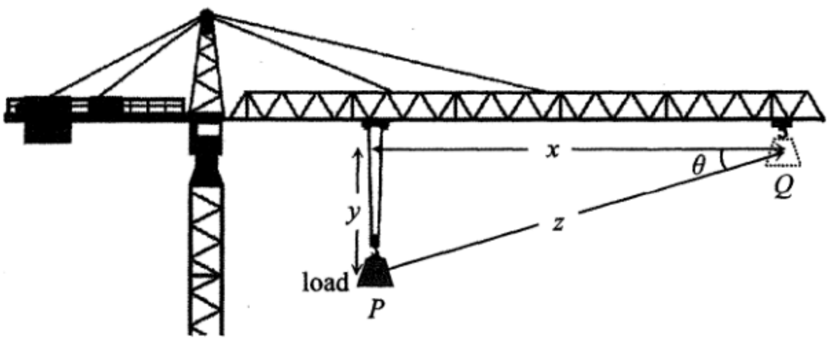


Which of the following correctly shows the variation of potential energy E_p and kinetic energy E_k with distance s that the block travelled up the incline ? Neglect air resistance.



2016 Q9

9. A crane moves a load of weight W steadily from point P to point Q as shown.



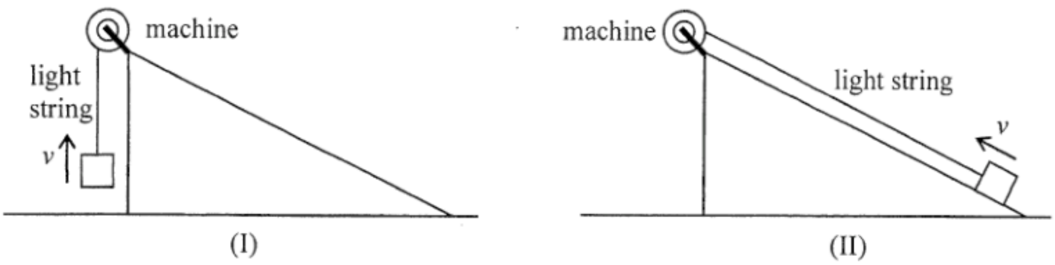
The work done on the load by the crane is

A. Wy.
B. $W(x+y)$.
C. Wz.
D. $Wz\cos\theta$.

2017 Q11

11. A machine is fixed at the top of a smooth inclined plane. Two methods, (I) and (II), are used to lift a block from the ground to the top of the inclined plane by the machine.

(I) Pull the block vertically upward at a uniform speed v.
(II) Pull the block up along the inclined plane at the same uniform speed v.



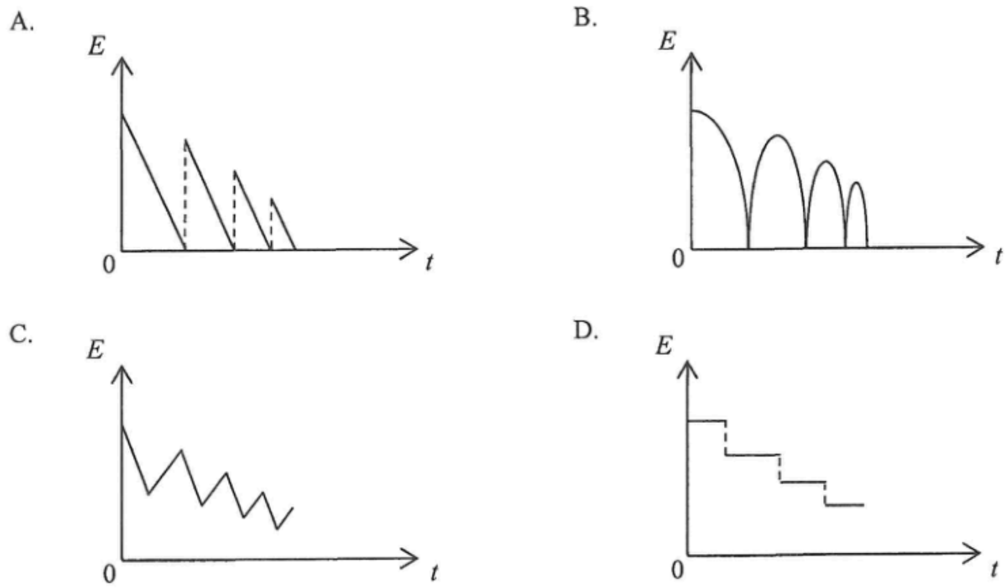
Which of the following statements correctly compare(s) the two methods ?

(1) The tension in the string is the same.
(2) The average output power of the machine is the same.
(3) The work done by the machine on the block is the same.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2026 Q9

9. A ball is released from a certain height. It rebounds from the ground several times before it comes to rest. Neglect air resistance. Which of the following graphs best represents the variation of the total mechanical energy E of the ball with time t ?



pp Q11

11.

A football player kicks a ball on the ground. The ball leaves the ground with speed v and hits the bar at X with a speed of 17 m s^{-1}. X is 2 m above the ground. Neglecting air resistance, what is the value of v ?

A. 15.8 m s^{-1}
B. 18.1 m s^{-1}
C. 19.0 m s^{-1}
D. 23.3 m s^{-1}